Arden Tsang

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EDUCATION

University of Cambridge Expected graduation: Jun 2027
MASt in Astrophysics Cambridge, UK

- Selected courses: QFT[x], GR[x], Black Holes[x], Field Theory in Cosmology[x], GWs & Numerical Relativity[x]

King's College London (KCL) Sep 2023 - Jul 2026
BSc Physics with Theoretical Physics London, UK

- **First Class Honours** every year, Ranked **Top 5%** of year. Overall weighted average: **79%** (final year **81%**)
- **King's Research Award.**
- **Layton Science Research Award.** "for the best promise of aptitude and genius for original scientific work."
- Relativistic Astrophysics[91], QM II[89], Condensed Matter[84], EM[83], Particle Physics[81], QM I[81], Medical Imaging[81], Optics[79], Math in Physics[79], Theoretical Physics[77] & Symmetry[76], Stat Mech[75].

RESEARCH & EXPERIENCE

CCP5 RESEARCH FELLOW; THEORETICAL & COMPUTATIONAL MATERIALS PHYSICS Dec 2025 – Present
KCL De Tomas Group London, UK

- **Awarded EPSRC-funded CCP5** (Condensed Matter Simulation Project) Bursary for full-time summer extension.
- Developed and characterised a hybrid PSO–DFT (GFN2-xTB/PBE+VV10) workflow for carbon-nanocluster structure prediction; benchmarked inertias, recovered C_{3-20} isomers and converged a C_{20} fullerene.
- Built HPC pipelines in Bash for GPU-accelerated optimiser tuning; authored READMEs for reproducibility. [Git]
- **Co-authoring** a manuscript on this work; submission targeted for end of July 2026.
- Curated benchmark datasets to validate against GAP-20/AIRSS, bridging ab initio accuracy with ML-IP scalability.
- Extending to compare simulated IR spectra against astrophysical molecular cloud observations.
- Extending PSO to oxygen/sulfur polymers, & initiating interpretability work on random-forest feature attribution; automated kinetic-state discovery via PySoftK. Supervised by Dr Carla De Tomas, Senior Lecturer in Net Zero.

PARTICLE PHYSICS & MACHINE LEARNING RESEARCH ASSISTANT Oct 2025 – Present
KCL Experimental Particle and Astroparticle group (EPAP) London, UK

- **Co-authored paper:** "Physics at the Edge: Benchmarking Quantisation Techniques and the EdgeTPU for Neutrino Interaction Recognition". (arXiv:2603.24607, submitted to EPJC)
- Presented poster at the AI for Science Conference, Alan Turing Institute, March 2026.
- Leading work (build, train, quantise) in GNNs (GATs & GCNs) & architecting ground up EGNNs to run on Edge.
- Running reconstruction and clustering with Pandora in LArSoft; writing ROOT feature-vector files as inputs for the group's set-transformer models. Developing proficiency in C++, LArSoft, and ROOT.
- Contributing to mechanistic interpretability work on set-transformer and graph-attention models.
- Trained and quantized CNNs (PTQ & QAT) with GENIE generated LArTPC neutrino datasets to support possible low latency, sustainable AI fast triggers. Supervised by Dr Stefano Vergani, Postdoctoral Research Associate.

PHOTOVOLTAICS UNDERGRADUATE RESEARCH ASSISTANT Sep 2025 – Jun 2026
KCL Functional Materials & Devices Laboratory London, UK

- Integrated CAD: Fusion360 & Inkscape with V4.75 CO_2 laser to nanostructure perovskite for spectral filtering.
- Investigated dichroic filters, perovskite for agrivoltaics; Supervised by Dr Jay Patel, Functional Photonics Lecturer.

ARTIFICIAL INTELLIGENCE UNDERGRADUATE SUMMER RESEARCH FELLOW Jul 2025 – Sep 2025
HKUST Industrial Engineering and Decision Analytics (IEDA) Hong Kong

- Designed and programmed scheduling algorithms for NPU architectures to optimize inference speed.
- Learned transformer theory & curated datasets to fine-tune LLMs with mlx to test physical derivation recovery.
- Interviewed published researchers on AI safety. Supervised by Dr Jiheng Zhang, Professor of IEDA. & Mathematics.

QUANTUM CHEMISTRY KING'S UNDERGRADUATE RESEARCH FELLOW (KURF) Jun 2025 – Jul 2025
KCL Physical Chemistry (Sanz) Group London, UK

- Successful identification of the lowest-energy isomer of limonene–$(SO_2)_2$ cluster via FT-CPMW spectroscopy.
- Employed Quantum Chemistry suite: DFT on the HPC via CLI; Spectral assignment, & NCI, NBO Chemical analysis.
- Awarded paid research fellowship; Supervised by Dr Maria Eugenia Sanz, Reader in Physical Chemistry.

ACTUARIAL INTERN Jul 2024 – Sep 2024
AIA Hong Kong

- Studied credit risk algorithms & created market projections, executed data analysis & crafted quarterly report.
- Documented linear algebra legacy code in Python and VBA from Excel to senior colleagues.

PUBLICATIONS, POSTERS & CONFERENCES

- Vergani, S.*, **Tsang, A.**, et al. Physics at the Edge: Benchmarking Quantisation Techniques and the Edge TPU for

- Neutrino Interaction Recognition. **AI for Science Conference, Alan Turing Institute**, London, UK. Mar 2026
- **Tsang, A.**, Kuchmina, S., de Tomas, C., et al. Simulating Carbon Nanoclusters using ML-IP & Particle Swarm Optimisation. **London Centre for Nanotechnology Symposium**; BSc Thesis Poster Session, London, UK, Apr 2026
- **Tsang, A.**, Sanz, M. E. Atmospheric Clusters: Structure and Interactions. Academic poster submitted for the King's Research Award, **King's Undergraduate Research Fellowship (KURF)**, London, UK, Oct 2025

POSITIONS OF RESPONSIBILITY

Cosmology Hackathon, KCL Department of Physics (London, UK) Jun 2026
- Built a ML pipeline to infer galaxy redshifts from 5-band SDSS photometry using GBTs and MLPs; RMSE ≈ 0.052.

Staff Writer, King's International Security Journal (KISJ) (London, UK) Sep 2025 – Present
- Sole author on AI's authoritarian tendencies, misalignment, collapse of information scarcity & macro-economy

Engineering, Philosophy Lead, KCL Rocketry Society (London, UK) Sep 2024 – Feb 2026
- Showcased technical exhibition & facilitated physical rocket construction utilizing Fusion360 for modeling.

Elected Head, SMUS Physics Club (Victoria, BC, Canada) Sep 2022 – Jun 2023
- Executed 15+ experiments, tested hypotheses, argued methods & results for theory; Hosted lectures on QM.

Co-Founder & Head, SMUS Engineering Club (Victoria, BC, Canada) Sep 2021 – Jun 2022
- Recruited 80+; Lectured; Directed bridge competition; Taught Inkscape to design & compete with printed car.

Physics Competitions; [IOP/IAPS], [CAP/IQC] (Swansea, UK); (Victoria, BC, Canada) Sep 2021 – Feb 2025
- **PLANCKS:** Winner of promotional event, competed & networked across UK universities in theory.
- **CaYPT:** Led team of 5 to design experiments, debated methods, results, & theories with schools across Canada.

STEM Tutor; Various Institutions (Victoria, BC, Canada); (London, UK) Sep 2021 – Present
- Instructed 15+ students in AP & IB Physics, AP Calc, Chem, Stat; across English & Cantonese, remote & in-person

COURSES & TRAINING

TECHNICAL AI SAFETY Jul 2026
BlueDot Impact Remote
- Alignment & RLHF, mech. interp., evaluations & red-teaming, AI control, & scalable oversight; capstone project.

FRONTIER AI GOVERNANCE Jul 2026
BlueDot Impact Remote
- Compute governance, safety standards, liability, & international coordination; analysis of AI policy proposals.

TECHNICAL SKILLS

- **Languages:** Native English, Native Cantonese, Fluent Mandarin, Limited Working Proficiency Spanish.
- **Programming & Computation:** Python (PySCF, gpu4pyscf, PyTorch, TensorFlow, Keras, MLX, SciPy, Pandas, Numpy, Matplotlib), Bash, MATLAB; Java; VBA; Swift;
- **HPC & Systems:** Linux/Unix CLI (MobaXterm, MacOS, Windows), Schedulers, GPU-accelerated workflows.
- **Software & Tools:** LaTeX; Git; Quantum Chemistry: PGOPHER, Pickett, Gaussian, Chimera, Multiwfn, OVITO, NetworkX; CAD: Fusion360, Inkscape, Prusa; Video editing/photoshop; Sketchup; Microsoft Office Suite.
- **Machine Learning & Numerical Methods:** LLM fine-tuning (MLX); GNNs; CNNs; Quantization: (Post-Training-Quantization, Quantization-Aware-Training), Stochastic Optimisation (PSO, Simulated Annealing); ODEs/PDEs (Mckenna's model, Taylor's, Tacoma); Packing (NPU); QM, MRI and ultrasound theory & visualisation.
- **Mathematical Physics:** Lie Algebras & Representation Theory; Analytical Mechanics; Quantum, Relativistic & Statistical Mechanics; Tensor Calculus; Axiomatic Set Theory.
- **Experimental Physics:** Optics & Photonics: Interferometry, Laser alignment, Spectroscopy, Polarimetry. Instrumentation: CO_2 lasers, Optical bench, Oscilloscopes, Helmholtz coils, NMR Spectrometers, Circuit design.
- **Leadership** abilities from society administrations & diverse collaborations across international environments.